

Subhav Kumar

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EDUCATION

B.Tech in Computer Science & Engineering 2023 – 2027
Indian Institute of Information Technology (IIIT), Raichur **CGPA: 8.99 / 10.00**
Relevant Coursework: Machine Learning, Probability & Statistics, Linear Algebra, Data Structures & Algorithms

EXPERIENCE

AI Research Intern — Wadhvani School of AI, IIT Madras May 2026 – Present
– Researching **Semantic Text Reconstruction from Embeddings** — developing decoder-based **deep learning** methods to recover natural language from dense vector representations using **transformer** architectures.
– Engineered evaluation framework in **Python** to benchmark reconstruction fidelity using **BLEU**, **cosine similarity**, and semantic equivalence metrics across multiple embedding models.
– Optimized distributed training pipelines on **HPC/GPU clusters**, improving throughput and memory efficiency for large-scale **NLP** experiments.

RESEARCH

Paper Accepted — IEEE ICCSE 2026 Budapest, Hungary
– “*PCA-based Defense Against Embedding Inversion and User History Leakage in Recommendation Systems.*”
– Proposed a **PCA-based dimensionality reduction** defense that mitigates **embedding inversion** and **membership inference** attacks on recommendation system user embeddings.
– Preserved recommendation quality while significantly reducing attacker reconstruction accuracy, demonstrating a practical **privacy-utility tradeoff** in **unsupervised learning** settings.

PROJECTS

CCTV Face Enhancement System [*OpenCV, Super-Resolution, Python*] Nov 2025 – Jan 2026
– Engineered a **computer vision** pipeline to detect, crop, and enhance low-resolution faces from CCTV footage using **super-resolution** models.
– Improved face recognition accuracy from **84% to 88%** (+4%) and image sharpness by **44%** (Laplacian variance: 165.85 → 238.98) with SSIM of **0.69**.
– Optimized inference to **330ms/face** on CPU, processing 75 face crops in 24.6s for near real-time enhancement.
–  [View Project Repository](#)

AgentReview AI — Multi-Agent PR Review System [*LangGraph, Gemini, FastAPI, Docker*] May'26 – June'26
– Built an **autonomous multi-agent system** with **4 specialized LLM agents** (Reviewer, Test Writer, Executor, Synthesizer) orchestrated via **LangGraph** state machine for AI-powered Pull Request review.
– Automated **code analysis**, **pytest generation**, and **sandboxed Docker execution**, producing GitHub-ready markdown reports with quality scores and actionable findings.
– Engineered **FastAPI** backend with **WebSocket** real-time updates and GitHub webhook integration for automated PR triggering.
–  [View Project Repository](#)

Semantic Text Reconstruction from Embeddings [*PyTorch, Transformers, NLP*] Ongoing
– Building a **transformer** pipeline in **PyTorch** to reconstruct natural language text from paragraph/document embeddings.
– Implemented custom training loops with **attention masking**, embedding injection, and **beam search** decoding.
– Evaluated output using **BLEU**, **cosine similarity**, and semantic equivalence metrics across multiple embedding models.
–  [View Project Repository](#)

TECHNICAL SKILLS

AI / ML	PyTorch, TensorFlow, Scikit-learn, Transformers, LangGraph, NumPy, Pandas, NLP, Computer Vision, LLM Fine-tuning (LoRA)
Languages	Python, C/C++, SQL, JavaScript, TypeScript
Mathematics	Probability, Statistics, Linear Algebra, Optimization, Dimensionality Reduction
Tools & Infra	Git, Docker, Linux, AWS, FastAPI, REST APIs, MongoDB, PostgreSQL, HPC/GPU Clusters

ACHIEVEMENTS

– **Amazon ML Challenge 2025:** Ranked **Top 1200** out of **80,000+** participants.
– **Competitive Programming:** Solved **400+ DSA problems** across LeetCode, GFG, and Codeforces. 
– **Hackathons:** Walmart Sparkathon  and Adobe India Hackathon  | **PRMO Qualified** (2019)

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 AI Foundations Associate
Master SQL and DBMS — Scaler

 [Verify](#)
 [Verify](#)